Customer Behavior & Purchase Analysis

Project Overview

Tools: SQL, Power BI, DAX, CSV

Project Type: Data Analytics / Business Intelligence

1. Business Problem

The business has customer purchase data, but it is difficult to understand customer buying patterns from raw data alone. The company needs a simple way to identify which products and categories generate more sales, how much customers spend, how subscriptions and shipping types relate to purchases, and where customer ratings differ.

Problem to solve: Convert raw customer transaction data into clear insights that can help the business understand customers and make better data-driven decisions.

2. Project Objective

The main objective is to analyze customer behavior using SQL and create an interactive Power BI dashboard so that important sales and customer patterns can be understood quickly.

The project focuses on revenue, purchase amount, product categories, customer ratings, subscriptions, shipping methods, gender and age groups.

3. Dataset

The dataset contains **3,900 customer records** with information such as age, gender, product, category, purchase amount, rating, subscription status, shipping type and previous purchases.

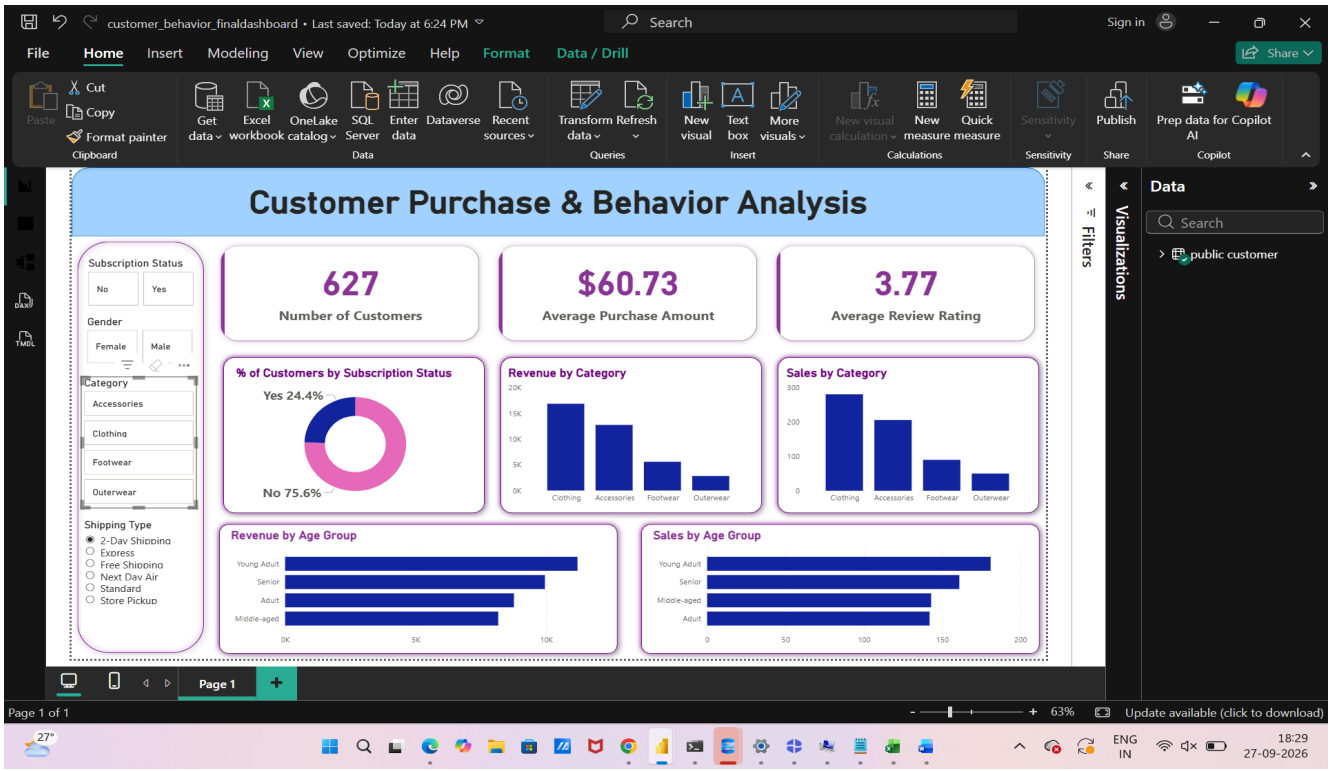
4. SQL Analysis

SQL was used to answer business questions such as revenue by gender, top-rated products, subscriber vs non-subscriber spending, shipping comparison, discount usage and top products within each category.

SQL concepts used: **SELECT, WHERE, GROUP BY, ORDER BY, SUM, AVG, COUNT, CASE WHEN, subqueries, CTE and ROW_NUMBER().**

5. Power BI Dashboard

The Power BI dashboard presents the analysis using KPI cards, charts and interactive slicers. Users can filter the dashboard by subscription status, gender, category and shipping type.



Main KPIs shown in the dashboard:

KPI	Value
Number of Customers	627*
Average Purchase Amount	$60.73*
Average Review Rating	3.77*

*Values shown are from the supplied dashboard screenshot/filter state.

6. Key Insights

- The dataset contains 3,900 customer transactions.
- Average purchase amount is $59.76.
- Average review rating is 3.75 out of 5.
- Clothing is the highest-revenue category in the supplied dataset.
- The dashboard allows interactive comparison of different customer and purchase segments.

7. Project Workflow

Raw Customer Data → SQL Analysis → Data Preparation → Power BI → Dashboard → Business Insights

8. What I Learned

• How to analyze customer data using SQL.
• How to create KPIs and charts in Power BI.
• How to use filters and slicers for interactive analysis.
• How to convert raw data into business insights.
• How SQL and Power BI can be combined in a Data Analyst project.

9. GitHub Files

• Power BI (.pbix) – Interactive dashboard
• CSV – Source dataset
• SQL – Analysis queries
• PNG – Dashboard screenshot
• PDF – Project overview